

Decision Tree

A simple example of Decision Tree Regression and Decision Tree Classifier with maths.

Decision Tree Regression:

Suppose we have a dataset with two features X and corresponding continuous target values Y:

X	Y
1	4
2	5
3	8
4	10
5	12

Let's build a Decision Tree Regression model to predict the target value Y based on feature X. We will use mean squared error (MSE) as the splitting criterion.

Step 1: Calculate the mean value of the target variable Y for the entire dataset:

$$\text{Mean_Y} = (4 + 5 + 8 + 10 + 12) / 5 = 39 / 5 = 7.8$$

Step 2: Calculate the MSE for the entire dataset:

$$\begin{aligned}\text{MSE} &= (1/5) * [(4-7.8)^2 + (5-7.8)^2 + (8-7.8)^2 + (10-7.8)^2 + (12-7.8)^2] \\ &= (1/5) * [13.44 + 8.84 + 0.04 + 4.84 + 18.36] \\ &= (1/5) * 45.52 \\ &= 9.104\end{aligned}$$

Step 3: Now, we find the best split for the feature X. Let's say we find the best split at X=3, which divides the dataset into two groups:

Group 1 ($X \leq 3$):

X	Y
1	4
2	5
3	8

Group 2 ($X > 3$):

X	Y
4	10
5	12

Step 4: Calculate the MSE for each group:

$$\begin{aligned}\text{MSE_Group1} &= (1/3) * [(4-5.67)^2 + (5-5.67)^2 + (8-5.67)^2] \\ &= (1/3) * [3.11 + 0.45 + 5.61] \\ &= (1/3) * 9.17 \\ &= 3.0567\end{aligned}$$

$$\begin{aligned}\text{MSE_Group2} &= (1/2) * [(10-11)^2 + (12-11)^2] \\ &= (1/2) * [1 + 1] \\ &= 1\end{aligned}$$

Step 5: Calculate the total MSE after the split:

$$\begin{aligned}\text{Total_MSE} &= (3/5) * \text{MSE_Group1} + (2/5) * \text{MSE_Group2} \\ &= (3/5) * 3.0567 + (2/5) * 1 \\ &= 2.0345 + 0.4 \\ &= 2.4345\end{aligned}$$

Step 6: Compare the Total_MSE with the MSE before the split. If the improvement is significant, continue splitting the nodes further. Otherwise, stop and create the final Decision Tree.

Decision Tree Classifier:

Suppose we have a dataset with two features X and corresponding binary class labels Y (0 or 1):

X	Y
1	0
2	0
3	1
4	1
5	1

Let's build a Decision Tree Classifier to predict the class label Y based on feature X. We will use Gini impurity as the splitting criterion.

Step 1: Calculate the Gini impurity for the entire dataset:

$$\begin{aligned}\text{Gini} &= 1 - (P(0)^2 + P(1)^2) \\ &= 1 - (2/5)^2 - (3/5)^2 \\ &= 1 - 4/25 - 9/25 \\ &= 1 - 13/25 \\ &= 12/25 \\ &= 0.48\end{aligned}$$

Step 2: Now, we find the best split for the feature X. Let's say we find the best split at X=3, which divides the dataset into two groups:

Group 1 ($X \leq 3$):

X	Y
1	0
2	0
3	1

Group 2 ($X > 3$):

X	Y
4	1
5	1

Step 3: Calculate the Gini impurity for each group:

$$\begin{aligned}\text{Gini_Group1} &= 1 - (P(0)^2 + P(1)^2) \\ &= 1 - (2/3)^2 - (1/3)^2 \\ &= 1 - 4/9 - 1/9 \\ &= 4/9 \\ &= 0.4444\end{aligned}$$

$$\begin{aligned}\text{Gini_Group2} &= 1 - (P(0)^2 + P(1)^2) \\ &= 1 - (0)^2 - (1)^2 \\ &= 1 - 0 - 1 \\ &= 0\end{aligned}$$

Step 4: Calculate the total Gini impurity after the split:

$$\begin{aligned}\text{Total_Gini} &= (3/5) * \text{Gini_Group1} + (2/5) * \text{Gini_Group2} \\ &= (3/5) * 0.4444 + (2/5) * 0 \\ &= 0.2666 + 0 \\ &= 0.2666\end{aligned}$$

Step 5: Compare the Total_Gini with the Gini before the split. If the improvement is significant, continue splitting the nodes further. Otherwise, stop and create the final Decision Tree.

The splitting process continues until a stopping criterion is met, such as reaching a maximum depth or a minimum number of samples in a node. The final Decision Tree will be used for prediction in the Decision Tree Classifier.